

Panel Data Econometrics

Chapter 4 – The fixed-effects model *revisiting the within estimator*

Reminder

In the previous chapter, we covered the GLS estimator.

The GLS estimator is BLUE, as long as all assumptions hold (especially A3).

As we do not know Ω , we estimate the FGLS estimator. This is **the random-effect model**. We discussed different methods to estimate $\hat{\Omega}$.

We saw that the GLS estimator always take a value between the OLS and the Within estimator.

Introduction

In this fourth chapter, we delve deeper into the within estimator and discuss the fixed effects model.

Part 1 – The philosophy behind the fixed effect model

Part 2 – The LSDV estimator

Part 3 – The within estimator

Part 4 – Choosing between random effects and fixed effects

Fixed effects: a change of paradigm

What are fixed effects ?

In the fixed effects model, we will consider that individual effects contain relevant information on individual specificities.

In contrast with the random-effects model where we brushed aside individual effects as the consequence of a shock, we are interested in identifying (or at least control for) individual effects associated to all our individual units.

- It could be interesting to compare individual units
- **Solves naturally the issues of endogeneity related to time-invariant individual heterogeneity.**

The LSDV estimator

The LSDV estimator

One way to estimate such a model would be to introduce among the explanatory variables a dummy variable for each of the individual units.

$$y = X\beta + D\gamma + \varepsilon$$

Where $\mathbf{D}$ is a $N \times N$ matrix including all individual dummy variables.

This is the Least Squares Dummy Variables (LSDV) estimator.

Note : To identify all parameters in γ , we need to remove the intercept from β (or we choose an individual as reference).

Fixed effects: a change of paradigm (bis)

Coming back to Frish-Waugh-Lovell

To understand how the introduction of fixed effects in the model can solve endogeneity issues, we can rely on the Frisch-Waugh-Lovell theorem.

Estimating the model presented above is the same as

- Regressing y on D , extracting the residuals y^*
- Regressing X on D , extracting the residuals X^*
- Regressing y^* on X^* . This gives us estimates of β that are « cleaned » from the impact of individual heterogeneity (i.e. all unobservables that corresponds to systematic differences between individual).

The LSDV estimator

The LSDV estimator

To estimate the model, the full rank hypothesis (A1) must be respected.

Therefore, it is impossible to identify both individual effects and the coefficients from any time-invariant variable.

In other words, a variable that takes always the same value over time for a given individual (e.g. education of the mother) cannot be introduced in the model.

The LSDV estimator

The LSDV estimator

Introducing N dummy variable naturally has a cost :

Degrees of freedom reduce to $(NT - K - N)$.

→ This could be an issue if the panel includes many individuals and few periods.

Also, keep in mind that individual effects are identified through the variability across the T observations per individual.

The LSDV estimator

The LSDV estimator - Application

We use the UnionWage dataset. This dataset has been used by Vella and Verbeek (1988) to investigate how union negotiations affect wages.

One issue arises: union membership is not exogeneous, and individuals select themselves into unions. Therefore, there could be an endogeneity issue: union membership could be correlated with individual characteristics that are omitted in the model.

The LSDV estimator

The LSDV estimator - Application

- See R code –
- We estimate the LSDV estimator.
- We compare it to the pooled estimator.

The within estimator

The within estimator

There is a more convenient (although exactly equivalent) way to estimate the fixed-effect model.

This is the within estimator that we covered in chapter 2.

Reminder: the within estimator consists in transforming the variables in deviations to the individual mean (within operator).

$$Wy = WX\beta + WD\gamma + W\varepsilon$$

With $WD = 0$

The within estimator

The within estimator

As we discussed in chapter 2, the within estimator also has $NT - N - K$ for its degrees of freedom.

The main benefits to estimate a fixed-effects model with the within estimator are computational, and practicality.

However, if we are interested in the values taken by individual effects, we will need to add an additional step to compute it.

The within estimator

The within estimator

From the within estimator, the individual effects can be calculated as :

$$\hat{\eta}_i = y_{i.} - x_{i.}'\hat{\beta}$$

If we want to center individual effects around 0, we need to first compute a general intercept (which would be the overall mean of the individual effects in the sample), and then compute individual deviations to this intercept.

$$\hat{\eta}_i = (y_{i.} - y_{..}) - (x_{i.} - x_{..})'\hat{\beta}$$

Application

The fixed effect model - application

- See R code –
- We estimate the within estimator and compare it to the LSDV.
- We retrieve individual effects manually.
- We center individual effects around 0.

Random or fixed ?

The trade-off between *random* and *within*

In the last two chapters, we covered the two main models deployed in the context of panel data analysis :

- The random-effect model
 - It relies on the FGLS estimator.
 - It has the best properties (precision), as long as there is no endogeneity issue.
 - If there is endogeneity, this estimator is not consistent.
- The fixed effect model
 - It relies on the within estimator.
 - It is consistent if there is endogeneity related to time-invariant characteristics.
 - It does not allow to investigate time-invariant covariates.

Random or fixed ?

The trade-off between *random* and *within*

1. If there is no correlation between the covariates and either η_i and ν_{it} : choose the random-effects model. It is consistent and has a lower variance than the fixed-effects model.
2. If there is correlation between some covariates and η_i (but not ν_{it}) : choose the fixed-effects model. It is consistent, whereas the random-effect model is not!
3. If there is endogeneity which is not limited to η_i , none of these estimators is consistent. Other estimators should be considered, in particular the Instrumental Variables estimator.

Random or fixed ?

The Hausman test

A common way to decide between random-effects model and fixed-effects model is to perform the Hausman test. We discussed a similar test already in the chapter related to endogeneity (Adv. Econometrics).

$$H = (\hat{\beta}^W - \hat{\beta}^{FGLS})' [Var(\hat{\beta}^W) - Var(\hat{\beta}^{FGLS})]^{-1} (\hat{\beta}^W - \hat{\beta}^{FGLS})$$

Under H_0 , $H \sim \chi_K^2$ (with K the number of parameters in the within estimation).

Random or fixed ?

Application

- See R code –
- We perform the Hausman test .
- We discuss the decision rule.